

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION
PHYSICS | SSC-I (9th Grade) | Units 1-6 (till 6.4) and 9 | Marking Report

Exam: exam-12-ch1-6-9 (Version 1) | Candidate: Seemab | Date: 3 August 2026 | Total Marks: 65

Marking basis: marked page by page against the exam paper and answer key, then every Section B and Section C calculation independently re-derived from the scan (kinematics acceleration/distance, net-force reasoning, layered-liquid pressure, snowmobile pressure, work done, equations-of-motion numerical, momentum conservation and KE loss, Hooke's law spring constant, and the falling-rock energy/air-resistance numerical). Section A checked letter by letter against the key at 400 DPI zoom after the first read returned 10/12, below Seemab's usual 11-12/12 baseline — re-verification confirmed both misses are genuine (Q.2 and Q.3), not a scan-reading artifact. One exam-paper data-integrity gap was found and is flagged separately below (Q.2(x) omits the force value the answer key assumes); Seemab was given full credit and not penalised for it. Final verified score: **56.5/65 (86.9%)**.

56.5 / 65
TOTAL SCORE

86.9%
PERCENTAGE

A
GRADE

10/12
MCQ

10/12

SECTION A: MCQS
83.3% - 2 misses

26.5/33

SECTION B: SHORT
80.3% - 3 questions cost marks

20/20

SECTION C: LONG
100% - flawless

Section A — Multiple Choice Questions (12 × 1 = 12 Marks)

#	Question (abbreviated)	Seemab	Correct	Result
1	Which is a base physical quantity?	B	B (Electric current)	CORRECT
2	Pascal (Pa), SI unit of pressure, is dimensionally equivalent to	A	C (kg/(m·s ²))	WRONG
3	Gradient of a distance-time graph gives the object's	B	C (Instantaneous speed)	WRONG
4	Ball dropped from rest, falls freely 7s, velocity at impact	D	D (70 m/s)	CORRECT
5	Strongest fundamental force (short range)	D	D (Strong force)	CORRECT
6	Jumper's feet push down, ground pushes back - illustrates	C	C (Newton's 3rd law)	CORRECT
7	Torque: 35N perpendicular, 0.5m from bolt	A	A (17.5 N·m)	CORRECT
8	Pencil balanced on tip, topples over - illustrates	B	B (Unstable)	CORRECT
9	Lines joining places of equal atmospheric pressure	B	B (Isobars)	CORRECT
10	Pressure at bottom of 7m deep water column	C	C (70,000 Pa)	CORRECT
11	Work done: 60N at 60°, moves crate 5m	C	C (150 J)	CORRECT
12	Branch of physics studying origin/future of the universe	B	B (Cosmology)	CORRECT

MCQ score: **10/12 (83.3%)**. Below Seemab's usual 11-12/12 baseline - re-verified at 400 DPI zoom (both "1.B 2.A 3.B 4.D 5.D" are unambiguous in her handwriting), so this is a genuine result, not a misread. Q.2 confuses pressure's dimension (kg/(m·s²)) with density's (kg/m³); Q.3 confuses a distance-time graph's gradient (instantaneous speed) with displacement. This ends Physics' run of largely-perfect MCQ sheets and is the first time Seemab has lost 2 MCQs on a single Physics paper. Physics MCQ record moves to 117/120 (97.5%, 3 errors: exam-10, and this exam's 2).

Section B — Short Answer Questions (11 × 3 = 33 Marks)

All 11 parts attempted, in full sequence with no gaps. Main chosen for 2(i), 2(ii), 2(iii), 2(vii), 2(ix), 2(x); OR chosen for 2(iv), 2(v), 2(vi), 2(viii), 2(xi).

Q.2(i): SI Base Quantities & Derived Units

Chose: Main 2/3

All seven SI base quantities listed correctly with units (length-metre-m, mass-kilogram-kg, time-seconds-s, temperature-Kelvin-K, electric current-Ampere-A, amount of substance-mole-mol, light intensity-candela-Cd). Velocity (m/s) and Force (kg·m/s²) units both correct. **-1: wrote "Area :- m³" - the correct SI unit for area is m² (length × length); m³ is the unit of volume, not area.**

Q.2(ii): Speed vs Velocity & Circular Track Reasoning

Chose: Main 3/3

Full comparison table correct: speed as distance/passage of time, scalar; velocity as displacement/passage of time, vector; symbols and formulas both given. Part (b) correctly answers: "No, the velocity is not constant because the direction of motion changes continuously while the car moves in circular path, even though it's speed remains constant." Matches the key's required reasoning exactly.

Q.2(iii): Equations of Motion - Delivery Van

Chose: Main 3/3

(a) $a = (32-8)/6 = 4 \text{ m/s}^2$ - correct. (b) Distance: an initial wrong attempt using $v=s/t$ (" $s = 24 \times 6 = 144\text{m}$ ") was self-corrected and crossed out, then the right kinematics formula was used: $s = ut + \frac{1}{2}at^2 = 48 + 72 = 120 \text{ m}$ - correct final answer, full marks. **Good habit: caught and struck through her own wrong first approach before submitting the final answer.**

Q.2(iv): Newton's Second Law - Mass Calculation

Chose: OR alternative 3/3

Newton's second law stated correctly: "The acceleration of an object is directly proportional to the magnitude of net force, in a same direction, is inversely protoproportional to the mass of an object." $F=ma$ given. $m = F/a = 45/5 = 9 \text{ kg}$ - correct.

Q.2(v): Net (Resultant) Force - Two People Pushing a Car

Chose: OR alternative 1/3

Correctly defines net force: "the net (resultant) force is overall acting on an object after adding all forces together, taking their directions into account." (1 mark). **-2: the specific numbers in the question (two people pushing a stalled car in the SAME direction with 18N and 25N) were never used anywhere. Instead, Seemab wrote three self-invented, unrelated examples with OPPOSING forces: "10N → [+] $\leftarrow$ 6N, Net force :- 4N", "9N → [+] $\leftarrow$ 5N, Net force :: 4N", "15N → [+] $\leftarrow$ 15N, Net force :: 0N. The object is balanced." None of these solve the set problem - the required answer (18+25=43N, "like parallel forces") never appears on the page. Single largest deduction on the paper.**

Q.2(vi): Terminal Velocity - Skydiver

Chose: OR alternative 3/3

"Terminal velocity is the maximum constant velocity speed of a falling object when air resistance becomes equal to it's weight." Then: "A skydriver's weight remains constant while air resistance increases as she falls faster. When air resistance equals her weight, the resultant force becomes zero, and she falls at a constant speed called terminal velocity." All three required marking points present (definition, initial acceleration reasoning, dynamic-equilibrium reasoning). Full marks.

Q.2(vii): Friction - Advantages, Disadvantages, Reduction	Chose: Main	3/3
Advantages: "friction holds the screw and nails in woods" + "friction b/w tyre and road helps to drive cars". Disadvantages: "it slows down moving objects and causes heating of moving parts in machinery" + "produce wear and tear". Method to reduce friction: "by polishing; if we polish the rough surfaces, they become smooth and friction is reduced." All three required components present and correct.		

Q.2(viii): Layered Liquid Pressure - Petrol on Water	Chose: OR alternative	3/3
(a) $P_1 = \rho gh = 700 \times 10 \times 2 = \mathbf{14,000 \text{ Pa}}$. (b) $P_2 = 1000 \times 10 \times 3 = 30,000 \text{ Pa}$; total $P_1 + P_2 = 14,000 + 30,000 = \mathbf{44,000 \text{ Pa}}$. Both correct, matches the key exactly.		

Q.2(ix): Pressure Definition & Snowmobile Numerical	Chose: Main	2.5/3
" <i>Pressure is defined as force per unit area. Pressure is represented by 'P'. It's equation is:- $P = F/A$ " then $P = F/A = 4500/0.25 = \mathbf{18,000 \text{ Pa}}$ - correct concept and correct final numeric answer. -0.5: the SI unit name "pascal" was never explicitly stated in the definition (only used as the "Pa" label on the final numeric answer) - the question's definition part specifically asks for the unit to be named, not just used.</i>		

Q.2(x): Work Done - Toy Wagon	Chose: Main	3/3 (paper defect)
" <i>Work is said to be done when force displaces in it's own direction.</i> formula:- $F.S$. $W = FS(\cos)(\theta)$. $W = f_6(0.77)$. $W = 4.62F$ " Exam-paper data-integrity gap, not a Seemab error: the printed question never states a force value (only $\cos 40^\circ = 0.77$ and displacement=6m are given), yet the answer key's model solution assumes $F=80\text{N}$. As printed, the question is not numerically solvable - Seemab correctly identified $W = Fd \cos \theta$, substituted the two values that ARE given, and simplified as far as the data allows ($W=4.62F$), the furthest any student could take this exact question. Full credit awarded; flagged for the exam-builder to add the missing "with a force of 80 N" phrase to future printings of this paper.		

Q.2(xi): Physics as an Interdisciplinary Field	Chose: OR alternative	0/3
The OR question asks for two examples of how physics principles apply to OTHER disciplines (e.g. Chemistry, Geology). Seemab wrote: " <i>Thermal Physics:- is the collective study of statistical mechanics, thermodynamics and kinetic theory of gases. Nuclear Physics:- is the study of the structure, properties and behaviour of atomic nuclei.</i> " -3: both "examples" given are sub-branches OF physics itself, not cases of physics being applied within a different discipline. None of the three required marking points (interdisciplinary framing, a chemistry-type example, a geology-type example) are addressed - this reads as a conceptual mix-up between "fields within physics" and "physics used inside other fields," rather than an incomplete answer. Largest single deduction on the paper by mark value alongside Q.2(v).		

Section B total: **26.5 / 33 (80.3%)** - 7 of 11 questions fully correct (plus one 2.5/3 near-miss on an unstated unit name); 3 questions cost marks, two from not fully answering the specific question set (Q.2(v), Q.2(xi)) rather than incomplete or incorrect working.

Section C — Long Answer Questions (4 × 5 = 20 Marks)

All 4 long questions attempted, every sub-part completed. Main chosen for 3(i) and 3(iv); OR chosen for 3(ii) and 3(iii).

Q.3(i): Derivation of $v=u+at$ & Van Kinematics Numerical	Chose: Main	5/5
(a) Correct derivation from $a=(v-u)/t \Rightarrow at=v-u \Rightarrow v=u+at$. (b) Given $u=7$, $v=27$, $s=136\text{m}$: (i) $v^2=u^2+2as \Rightarrow 729=49+272a \Rightarrow a=\mathbf{2.5 \text{ m/s}^2}$. (ii) $v=u+at \Rightarrow 27=7+2.5t \Rightarrow t=\mathbf{8 \text{ s}}$. Both correct, matches key exactly.		

Q.3(ii): Conservation of Momentum - Car/Truck Collision	Chose: OR alternative	5/5
(a) Law of conservation of momentum stated correctly: "the total momentum of an isolated system remains constant if no external force acts on it." (b) $m_1v_1+m_2v_2=(m_1+m_2)v_f \Rightarrow 1000(18)+1500(-6)=2500v_f \Rightarrow v_f=\mathbf{3.6 \text{ m/s}}$. (c) $KE_i = \frac{1}{2}(1000)(18)^2+\frac{1}{2}(1500)(6)^2 = 189,000 \text{ J}$; $KE_f = \frac{1}{2}(2500)(3.6)^2 = 16,200 \text{ J}$; loss = 172,800 J . All three parts correct, matches key exactly.		

Q.3(iii): Hooke's Law & Spring Constant	Chose: OR alternative	5/5
Hooke's law: "within the elastic limit, the extension of a spring is directly proportional to the applied force." $F=kx$ given; spring constant defined as "the force required to produce unit extension in a spring," SI unit N/m. (i) $k=F/x=10/0.05=\mathbf{200 \text{ N/m}}$. (ii) Max load = $kx=200 \times 0.12=\mathbf{24 \text{ N}}$. All correct.		

Q.3(iv): Conservation of Energy - Falling Rock	Chose: Main	5/5
(a) "Energy can neither be created nor be destroyed; it can only be transformed from one form to another." (b)(i) $PE=mgh=6 \times 10 \times 45=\mathbf{2700 \text{ J}}$. (ii) $KE=PE=2700 \text{ J} \Rightarrow v^2=2700/3=900 \Rightarrow v=\mathbf{30 \text{ m/s}}$ (she began a second, equally valid route - " $2700 \times 2=6v^2$ ", which also resolves to $v^2=900$ - then crossed it out mid-way and switched to the quicker fraction-simplification method shown above; not an error, just an abandoned correct path). (c) Actual KE at 24 m/s = 1728 J; work done against air resistance = $2700-1728=\mathbf{972 \text{ J}}$. All correct, matches key exactly.		

Section C total: **20 / 20 (100%)** - a flawless section, all four long questions fully correct including two self-corrected calculations that were caught and fixed before the final answer.

Complete Deduction Log (reconciles to headline score)			
Question	Awarded	Lost	Reason
MCQ Q.2 - SI unit of pressure	0/1	-1	Chose A " kg/m^3 " (density's dimension); correct is C " $\text{kg}/(\text{m}\cdot\text{s}^2)$ " (force ÷ area).
MCQ Q.3 - Distance-time graph gradient	0/1	-1	Chose B "Displacement"; correct is C "Instantaneous speed" (the gradient of a distance-time graph gives speed, not displacement).
Q.2(i) Main - SI base quantities	2/3	-1	Wrote "Area :- m^3 " instead of the correct m^2 ; all seven base quantities and both velocity/force units correct.
Q.2(v) OR - Net force, two people pushing a car	1/3	-2	Correct definition given, but the actual numbers in the question ($18\text{N} + 25\text{N}$, same direction) were never used; three self-invented, irrelevant examples with opposing forces were written instead.
Q.2(ix) Main - Pressure definition & SI unit	2.5/3	-0.5	Correct definition, equation, and final numeric answer (18,000 Pa), but the SI unit's name "pascal" was never explicitly stated.
Q.2(xi) OR - Physics as an interdisciplinary field	0/3	-3	Defined "Thermal Physics" and "Nuclear Physics" (both sub-branches of physics itself) instead of giving examples of physics applied within OTHER disciplines, as required.
Total deductions			65 - 8.5 = 56.5/65 confirmed. Every other Section A, B, and C question scored full marks. Section C is a flawless 20/20.

Exam-paper data-integrity gap (flagged for the exam-builder, not a mark against Seemab): Q.2(x) ("Define work done. A child pulls a toy wagon with a rope at 40° above horizontal, moving it 6m. Taking $\cos 40^\circ = 0.77$, calculate the work done.") never states a force value in the printed question, but the answer key's model solution assumes $F=80\text{N}$. As printed, the question cannot be solved to a final number. Seemab was given full credit for taking the working as far as the printed data allows.

Strengths in this attempt

- **Flawless Section C (20/20, 100%)** – all four long questions fully correct: the $v=u+at$ derivation and van kinematics, momentum conservation with a fully correct 3-part KE-loss calculation, Hooke's law spring-constant derivation and numerical, and the falling-rock energy/air-resistance numerical.
- **Self-correction habit** – Q.2(iii) caught and struck through a genuinely wrong first attempt (the $v=s/t$ shortcut) before switching to the correct kinematics formula; Q.3(iv) shows a different, more mature habit – she started a second valid algebraic route (" $2700 \times 2 = 6v^2$ ", which also gives $v^2=900$) and abandoned it mid-way for a quicker simplification, rather than fixing an error.
- Layered-liquid pressure (petrol on water) and the momentum/KE-loss collision numerical both fully correct across every sub-part with complete, clearly laid-out working.
- Terminal velocity and speed-vs-velocity reasoning both explained with the full causal "why," not just the definitions.

Priority revision targets

- **Answer the specific numbers given, not a self-invented version (–2):** Q.2(v) asked for the net force of 18N and 25N pushing the same way; Seemab wrote a correct general definition but then solved three made-up examples of her own instead of the set numbers. Drill fix: after writing any definition, re-read the question's own numbers and plug them in before moving on.
- **Branches OF physics vs physics APPLIED to other fields (–3):** Q.2(xi)OR asked for examples of physics used inside other disciplines (e.g. chemistry, geology); "Thermal Physics" and "Nuclear Physics" are sub-fields of physics itself, not interdisciplinary examples. Single largest deduction on the paper.
- **SI unit precision:** wrote "Area = m^3 " (a volume unit) instead of m^2 , right after correctly listing length's own unit as "m." Quick flash-card fix: Area = length $\times$ length = $m \times m = m^2$ (never m^3 , which needs three lengths multiplied).
- **Naming the unit, not just using it:** Q.2(ix)'s definition never said "SI unit: pascal" in words, even though "Pa" appeared correctly in the final numeric answer.
- Two Section A dimensional/graph-concept MCQs missed together on one paper – first time this has happened for Physics; worth a quick refresher on unit dimensional analysis (pressure vs density) and graph-gradient meanings (distance-time vs speed-time).
- **Spelling/wording watchlist (no marks lost this time):** "acceleration" instead of "acceleraton" recurs on pages 3, 4 and 9; "defination" instead of "definition" (page 9); "it's" used throughout in place of the possessive "its"; consistently spells "skydiver" as "skydriver" (e.g. Q.2(vi)'s "a skydriver's weight remains constant"). Q.2(x)'s work-done definition was also loosely worded – "work is said to be done when force displaces in it's own direction" – but was rescued by the correct formula $W = FS \cos \theta$ that followed it. None of these cost marks on this paper, but worth tidying up before a stricter examiner starts docking for them.

Physics Exam Comparison

Exam	Date	Score	%	MCQ	Sec B	Sec C	Grade
exam-01-ch1-2	7 Apr 2026	63.5/65	97.7%	12/12	32.5/33	19/20	A+
exam-02-ch3-4-9	10 Apr 2026	62/65	95.4%	12/12	30.5/33	19.5/20	A+
exam-03-ch5	18 May 2026	65/65	100%	12/12	33/33	20/20	A+
exam-04-ch6	22 Jun 2026	55.5/65	85.4%	12/12	29/33	14.5/20	A
exam-07-ch1-6-9	25 Jun 2026	58.5/65	90.0%	12/12	31.5/33	15/20	A+
exam-08-ch1-6-9	2 Jul 2026	61/65	93.8%	12/12	30.5/33	18.5/20	A+
exam-09-ch1-6-9	7 Jul 2026	64.5/65	99.2%	12/12	32.5/33	20/20	A+
exam-10-ch1-6-9	16 Jul 2026	64/65	98.5%	11/12	33/33	20/20	A+
exam-11-ch1-6-9	23 Jul 2026	60.5/65	93.1%	12/12	29.5/33	19/20	A+
exam-12-ch1-6-9 (this)	3 Aug 2026	56.5/65	86.9%	10/12	26.5/33	20/20	A

exam-12 (56.5/65, 86.9%) is Physics' lowest score since exam-04-ch6 (85.4%), driven by Section A's first-ever 2-MCQ miss and two Section B questions (2(v), 2(xi)) that answered a different question than the one set, rather than incomplete/incorrect working – Section C, in contrast, is a flawless 20/20. Physics' subject average adjusts to 94.0% (611.0/650, 10 exams), and the subject drops from #1 to #3, now behind both Chemistry (94.8%) and Math (94.3%).